



CS 309: Information and Database Systems

Course Code	CS 309
Course Name	Information and Database Systems
L-T-P-C	3-0-2-4
Prerequisites	CS 207 Applied Databases Practicum
Intended for	UG
Distribution	Compulsory for CSE; CS elective for EE and ME
Semester	6th
One-time approval	recommended by the 9th BoA (Oct 2015)

Course Contents

- **Introduction** [6 Lectures]
 - Information Modeling: background, approaches, information system lifecycle.
 - Four information levels: conceptual, logical, physical, external
 - Conceptual Schema Design Procedure : facts, constraints, roles, value, set comparison, final checks
- **Relational Database Design** [6 Lectures]
 - Overview of ER, Barker notation, mapping from ORM to ER
 - Overview of UML, mapping from ORM to UML
 - Relational schemas, functional dependencies, Normal forms
- **Data Manipulation with SQL** [9 Lectures]
 - Relational Algebra
 - SQL : Basic operations, Joins, Nested and correlated queries, views, Triggers
 - Embedded SQL and database application development
- **Transactions** [9 Lectures]
 - ACID properties
 - Concurrency Control Techniques
 - Recovery Techniques
- **Principles of query processing** [3 Lectures]
 - Indexes
 - Query plans and operators
 - Cost-based query optimization
- **Data storage** [3 Lectures]
 - Databases Vs. FileSystems (Google FileSystem, Hadoop Distributed FileSystem)
- **Scalable data processing** [6 Lectures]
 - MapReduce and introduction to systems based on MapReduce (Hadoop)
 - Introduction to Scalable key-value stores (Amazon Dynamo, Google BigTable, HBase)

Text Books

1. Terry Halpin and Tony Morgan, **Information Modeling and Relational Databases**, 2nd Edition ((The Morgan Kaufmann Series in Data Management Systems) by

References

1. Elmasri, Ramez and Navathe, **Fundamentals of Database Systems**, 6th Edition, Shamkant
2. Raghu Ramakrishnan and Johannes Gehrke, **Database Management Systems**, 3rd Edition, McGraw- Hill
3. Assorted Readings from web resources like Yahoos Hadoop Tutorial, Googles Big Table etc.